Relative Loan Pricing as an Early-Warning Signal of Non-Performing Loans

Endogenous screening under borrower opacity · Track B

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github.com/fgalarzac/ai-project

The information problem

- Loan prices are observed when credit is originated.
- Portfolio risk becomes visible only after loans mature.
- A supervisor therefore needs an early-warning signal.

Outcome to forecast

$$\text{NPL} = \frac{\text{non-performing balance}}{\text{total outstanding balance}}$$

The exact non-performing classification follows the data provider's definition.

A high lending rate can reflect **riskier applicants** or **stronger bank screening**. Which signal wins?

When does a higher loan spread relative to peer banks predict a higher future NPL ratio—and when can it predict a lower one?

Track B adaptation

The thesis proposed selection minus screening in reduced form. This project adds the bank's screening choice, derives the sign from a first-order condition, and identifies an opacity threshold.

Interpretation boundary

This is a forecasting mechanism conditional on an observed price, not the causal effect of forcing a bank to raise its lending rate.

1 · Constructing the price signal

For bank j, credit segment s, and month t:

$$s_{jst} = i_{jst} - r_t, \qquad D_{jst} = s_{jst} - \text{median}_{k \neq j}\{s_{kst}\}.$$

i_{jst}	bank lending rate
r_t	policy rate
s_{jst}	spread over policy
D_{jst}	deviation from peer median
k	another bank in the peer group

Illustrative example

Lending rate: 18% Policy rate: 5%
Bank spread: 13% Peer median: 11%

$D = 13\% - 11\% = 2$ percentage points

Comparison is within product, maturity, currency, and month.

1 · Why the sign is ambiguous

selection channel

A bank priced above peers loses comparison shoppers and may be left with a riskier applicant pool.

$$\frac{\partial q^0}{\partial D} = a(\kappa) > 0$$

Higher $D \Rightarrow$ higher pre-screen risk.

screening channel

A performing loan is more valuable at a higher price, so preventing default through screening becomes more valuable.

$$\frac{\partial e^*}{\partial D} > 0$$

Higher $D \Rightarrow$ more screening.

Borrower opacity changes the relative strength of these channels.

2 · Timing: who chooses what, and when?

1. A borrower segment with opacity κ is given.
2. The bank posts relative spread D and faces potential volume $L(D, \kappa)$.
3. Price sorting creates an applicant pool with risk $q^0(D, \kappa)$.
4. After observing the pool, the bank chooses screening effort e.
5. Screening produces booked risk $q(D, e, \kappa)$.
6. At horizon h, the portfolio NPL ratio is observed.

Why this order matters

D is predetermined inside the screening subgame, so the bank's effort response can be solved cleanly and then mapped to future portfolio risk.

2 · Notation I: state, choices, and probabilities

Symbol	**Meaning**
$\kappa \in [0,1]$	opacity of the borrower segment; higher means harder to compare
$D \in \mathcal{D}$	relative spread chosen by the bank; $\mathcal{D}$ is compact
$L(D,\kappa)$	potential loan volume, with $L_D < 0$
$\bar{q}$	baseline default probability
$q^0(D,\kappa)$	default probability before screening
$e \in [0,\bar{e}]$	screening effort and its maximum feasible level
$q(D,e,\kappa)$	default probability after screening

κ is an economic characteristic of a segment; it is not the peer-bank index k used in the median.

2 · Notation II: technology, payoffs, and result

Symbol	**Meaning**
$a(\kappa) > 0$	sensitivity of pre-screen risk to the relative spread
$g(\kappa) > 0$	reduction in default probability from one unit of effort
$c(\kappa) > 0$	convex screening-cost shifter
$R(D) = \bar{R} + D$	payoff if the loan performs
F	funding and operating cost independent of effort
$h(\kappa) = g(\kappa)^2/c(\kappa)$	screening leverage; not effort itself
$\beta(\kappa)$	optimized risk slope $\partial q^*/\partial D$
κ^*	opacity at which $\beta(\kappa^*) = 0$

Rates and probabilities are written as decimals; model parameters are restricted so all probabilities remain in $(0, 1)$.

2 · Step 1: price sorting forms the pool

$$\boxed{q^0(D,\kappa) = \bar{q} + a(\kappa)D}, \qquad a(\kappa) > 0.$$

- q^0: probability of default among applicants *before* the bank screens them.
- $\bar{q}$: baseline risk when the bank is priced at the peer median, $D = 0$.
- $a(\kappa)$: how strongly the composition of applicants changes when the bank moves above or below peers.

Maintained selection restriction

$a'(\kappa) < 0$: price sorting is weaker in more opaque segments because applicants are assumed to compare offers less effectively.

2 · Step 2: screening changes booked risk

$$\boxed{q(D, e, \kappa) = q^0(D, \kappa) - g(\kappa)e}$$

Technology

- e: investigation, verification, and underwriting effort.
- $g(\kappa)e$: reduction in default probability.
- $g > 0$: more effort improves booked quality.

Cost

$$C(e, \kappa) = \frac{c(\kappa)}{2}e^2$$

Convexity means each additional unit of effort is more expensive than the previous one.

Interiority requires $0 < e^*(D, \kappa) < \bar{e}$; the code verifies it over the illustrative parameter grid.

$$\max_{D\in\mathcal{D}} L(D,\kappa)\left\{\max_{e\in[0,\bar{e}]} \underbrace{[1-q^0(D,\kappa)+g(\kappa)e](\bar{R}+D)}_{\text{expected performing-loan payoff}} - \underbrace{F}_{\text{other cost}} - \underbrace{\frac{c(\kappa)}{2}e^2}_{\text{screening cost}}\right\}.$$

- The outer problem chooses the relative price and loan volume.
- The inner problem chooses screening after the pool is observed.
- A compact $\mathcal{D}$ and continuity give a first-stage solution.

Scope of the proposition

I solve the inner problem exactly. The result is the conditional quality map $q^*(D,\kappa)$, not a closed-form oligopoly price D^*.

Marginal benefit = marginal cost:

$$\frac{\partial \pi}{\partial e} = \underbrace{g(\kappa)(\bar{R} + D)}_{\text{benefit of one more effort unit}} - \underbrace{c(\kappa)e}_{\text{marginal effort cost}} = 0.$$

$$\frac{\partial^2 \pi}{\partial e^2} = -c(\kappa) < 0 \quad \Rightarrow \quad \boxed{e^*(D, \kappa) = \frac{g(\kappa)}{c(\kappa)}(\bar{R} + D)}.$$

Economic comparative static

$$\frac{\partial e^*}{\partial D} = \frac{g(\kappa)}{c(\kappa)} > 0.$$

A larger payoff from a performing loan makes screening more valuable.

3 · Substitution reveals the two forces

Substitute optimal effort into the risk technology:

$$q^*(D,\kappa) = \bar{q} + a(\kappa)D - \underbrace{\frac{g(\kappa)^2}{c(\kappa)}}_{h(\kappa)}(\bar{R} + D).$$

Differentiate with respect to the observed relative spread:

$$\beta(\kappa) = \frac{\partial q^*}{\partial D} = \underbrace{a(\kappa)}_{\text{selection: pool worsens}} - \underbrace{h(\kappa)}_{\text{screening: quality improves}}.$$

$\beta > 0$: high price predicts more risk. $\beta < 0$: high price predicts less risk.

3 · Candidate Proposition 1: an opacity threshold

Assume interiority and feasible probabilities, plus

$$a, h \text{ continuous}, \quad a'(\kappa) < 0, \quad h'(\kappa) > 0, \quad a(0) > h(0), \quad a(1) < h(1).$$

Then one and only one $\kappa^* \in (0,1)$ solves $a(\kappa^*) = h(\kappa^*)$, and

$$\beta(\kappa) \begin{cases} > 0, & \kappa < \kappa^*, \\ = 0, & \kappa = \kappa^*, \\ < 0, & \kappa > \kappa^*. \end{cases} \qquad \frac{\partial^2 q^*}{\partial D \partial \kappa} = a' - h' < 0.$$

Proof idea: continuity plus endpoint crossing gives existence; strict monotonicity of $a - h$ gives uniqueness.

3 · The key restriction—and the important correction

$$h(\kappa) = \frac{g(\kappa)^2}{c(\kappa)}$$

Log differentiation gives

$$\frac{h'}{h} = 2\frac{g'}{g} - \frac{c'}{c}.$$

What is required

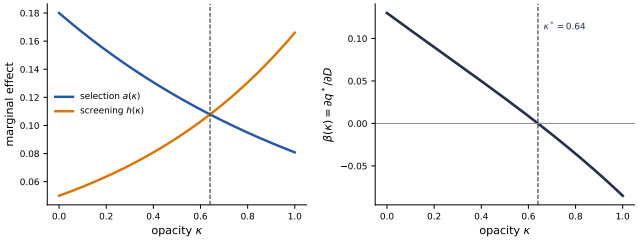
$h' > 0$ only if the marginal information gain g grows fast enough to dominate any rise in screening cost c.

What is *not* true

Higher screening cost by itself does not generate a safer high-price bank. It lowers h and makes screening less likely to dominate selection.

This visible, falsifiable condition is the conceptual improvement over the thesis's reduced-form statement.

3 · Illustrative calibration: not empirical evidence



- Blue: selection $a(\kappa)$ falls.
- Orange: leverage $h(\kappa)$ rises.
- Their crossing creates $\kappa^* \approx 0.64$.
- Right panel: $\beta = a - h$ changes sign once.

Purpose: verify logic, bounds, and a non-empty parameter region.

$$\boxed{\mathrm{NPL}_{js,t+h} = \alpha_h + \rho_h q^*_{jst} + u_{js,t+h}}, \qquad \rho_h > 0.$$

- q^*_{jst}: theoretical default probability attached to new lending at origination.
- $\mathrm{NPL}_{js,t+h}$: later stock ratio for the portfolio cell.
- α_h: horizon-specific baseline; $u_{js,t+h}$: other shocks.
- $\rho_h > 0$: riskier originations weakly worsen the later portfolio.

Testable implication

The predictive slope has sign $\mathrm{sign}[\rho_h \beta(\kappa)]$. The empirical test asks whether D improves out-of-sample NPL-ratio forecasts by opacity group and horizon.

4 · Where the contribution sits in the literature

Baseline	What it contributes	What this project adds
Varian (1980)	Shoppers and captive buyers generate price dispersion.	Adds bank screening and loan default risk.
Broecker (1990)	Rate competition interacts with imperfect credit tests.	Makes screening a continuous effort choice.
Ruckes (2004)	Screening intensity responds to prior borrower quality.	Studies the sign of a relative-price signal across opacity.
Dell'Ariccia and Marquez (2006)	Information asymmetry generates screening or pooling regimes.	Derives a continuous $a(\kappa) - h(\kappa)$ decomposition and threshold.

Narrow novelty claim: relative-price sorting + endogenous screening + a condition for the sign of the forecasting signal.

5 · Execution plan, risks, and final takeaway

Execution

1. Prove the threshold and a general-$R(D)$ robustness result.
2. Map the full admissible parameter region in simulation.
3. Compare rolling out-of-sample NPL-ratio forecasts.
4. Formalize the frozen theorem in Lean.

Risks and disciplined fallbacks

- If $h' > 0$ fails, report the $a > h$, $a = h$, and $a < h$ regions without forcing a threshold.
- If pricing confounds interpretation, retain a predictive design with bank and segment-by-month controls.

Takeaway

The contribution is not "high rates are safe." It is the explicit condition under which a relative price switches from a risk signal to a screening signal.